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SECTION: BSCpE 3-4	Professor: Dr. Lutzer U. Reyes
INSTRUCTIONS: • Perform the laboratory activity below, which involves creating a Computer Network Topology in Packet Tracer, assigning device addresses in the address table, and following the provided procedures. • Screenshots of each activity performed must be captured and included in the final submission. The screenshots must be placed after or below the corresponding number in the step-by-step procedures.	

Packet Tracer - Implement EtherChannel

Objectives

Part 1: Build the network

Part 2: Configure EtherChannel

Background

You have been assigned the task of designing an EtherChannel implementation for a company that wants to improve the performance of their switch trunk links. You will try several different ways of implementing the EtherChannel links in order to evaluate which is the best for the company. You will build the topology, configure trunk ports, and implement LACP and PAgP EtherChannels.

Instructions

Part 1: Build the network.

Use the table below to build the switch topology.

Step 1: Obtain the devices that are required.

- Click the **Network Devices** icon in the bottom tool bar.
- Click the **Switches** entry in the submenu.
- Locate the **2960** switch icon. Click and drag the icon for the 2960 switch into the topology area.
- Repeat the step above so that there are **three** 2960 switches in the topology area.

- e. Arrange the devices into a layout that you can work with by clicking and dragging.

Step 2: Name the devices.

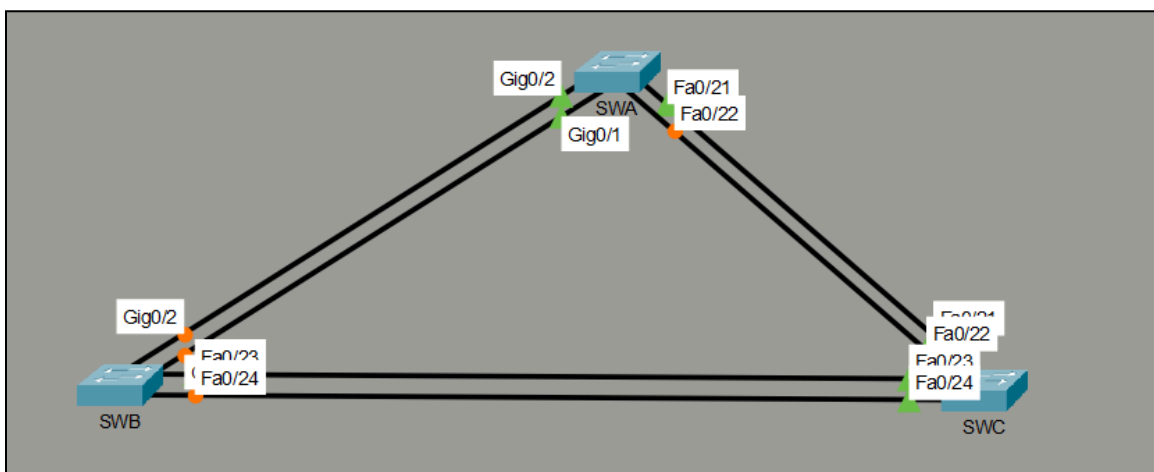
The devices have default names that you will need to change. You will name the devices **SWA**, **SWB**, and **SWC**. You are changing the display names of the devices. This is the text label that appears below each device. It is **not** the host name. Your display names must match the names that are given in this step **exactly**. If a display name does not match, you will not be scored for your device configuration.

- a. Click the device display name that is below the device icon. A text field should appear with a flashing insertion point. If the configuration window for the device appears, close it and try again by clicking a little further away from the device icon.
- b. Replace the current display name with the appropriate display name.
- c. Repeat until all devices are named.

Step 3: Connect the devices.

- a. Click the orange lightning bolt Connections icon in the bottom toolbar.
- b. Locate the Ethernet straight-through cable icon. It looks like a solid black diagonal line.
- c. To connect the device, click the Ethernet straight-through cable icon and then click the first device that you want to connect. Select the correct port and then click the second device. Select the correct port and the devices will be connected.
- d. Connect the devices as specified in the table below.

Port Channel	Devices	Port Connections	Type
1	SWA to SWB	G0/1 to G0/1	PAgP
		G0/2 to G0/2	
2	SWA to SWC	F0/21 to F0/21	LACP
		F0/22 to F0/22	
3	SWB to SWC	F0/23 to F0/23	LACP
		F0/24 to F0/24	



Part 2: Configure EtherChannel

On each switch, configure the ports that will be used in the Port Channels as static trunk ports.

Step 1: Configure a PAgP EtherChannel.

Follow the procedure that was used in previous activities to configure Port Channel 1 as a PAgP EtherChannel between SWA and SWB. Both sides should negotiate the EtherChannel.

```
SWA(config)#int range g0/1-2
SWA(config-if-range)#channel-group 1 mode desirable
%EC-5-L3DONTBNDL2: Gig0/1 suspended: PAGP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: PAGP currently not enabled on the remote port.
SWA(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

SWA(config-if-range)#exit
SWA(config)#interface port-channel 1
SWA(config-if)#switch
SWA(config-if)#switchport mode trunk
SWA(config-if)#end
SWA#
%SYS-5-CONFIG_I: Configured from console by console

SWA#write memory
```

```
SWB#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SWB(config)#interface range g0/1 - 2
SWB(config-if-range)#channel-group 1 mode desirable
%EC-5-L3DONTBNDL2: Gig0/1 suspended: PAGP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Gig0/2 suspended: PAGP currently not enabled on the remote port.
SWB(config-if-range)#
Creating a port-channel interface Port-channel 1

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface Port-channel1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to up

SWB(config-if-range)#exit
SWB(config)#interface port-channel 1
SWB(config-if)#switchport mode trunk
SWB(config-if)#
%LINK-3-UPDOWN: Interface Port-channel1, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel1, changed state to down

SWB(config-if)#end
SWB#
%SYS-5-CONFIG_I: Configured from console by console

SWB#write memory
```

Step 2: Configure a LACP EtherChannel.

Configure Port Channel 2 as an LACP channel between SWA and SWC. Both sides should negotiate the EtherChannel.

```
SWA#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SWA(config)#int range f0/21-22
SWA(config-if-range)#channel-group 2 mode active
%EC-5-L3DONTBNDL2: Fa0/21 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/22 suspended: LACP currently not enabled on the remote port.
SWA(config-if-range)#
Creating a port-channel interface Port-channel 2

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to up

SWA(config-if-range)#exit
SWA(config)#int port-channel 2
SWA(config-if)#switch
SWA(config-if)#switchport mode trunk
SWA(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to up

SWA(config-if)#end
SWA#
%SYS-5-CONFIG_I: Configured from console by console

SWA#write memory
```

```
SWC#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SWC(config)#interface range f0/21 - 22
SWC(config-if-range)#channel-group 2 mode active
%EC-5-L3DONTBNDL2: Fa0/21 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/22 suspended: LACP currently not enabled on the remote port.
SWC(config-if-range)#
Creating a port-channel interface Port-channel 2

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to up
%LINK-5-CHANGED: Interface Port-channel2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

SWC(config-if-range)#exit
SWC(config)#int port-channel 2
SWC(config-if)#switch
SWC(config-if)#switchport mode trunk
SWC(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/21, changed state to up
%LINK-3-UPDOWN: Interface Port-channel2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/22, changed state to up
%LINK-5-CHANGED: Interface Port-channel2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel2, changed state to up

SWC(config-if)#end
SWC#
%SYS-5-CONFIG_I: Configured from console by console

SWC#write memory
```

Step 3: Configure a Backup LACP EtherChannel

Configure Port Channel 3 channel as an LACP channel between SWB and SWC. In this case, SWC initiates negotiation with SWB. SWB does not initiate negotiation of the channel.

```
SWB(config)#interface range f0/23 - 24
SWB(config-if-range)#channel-group 3 mode passive
SWB(config-if-range)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

SWB(config-if-range)#exit
SWB(config)#int port-channel 3
SWB(config-if)#switch
SWB(config-if)#switchport mode trunk
SWB(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up

SWB(config-if)#end
SWB#
%SYS-5-CONFIG_I: Configured from console by console

SWB#write memory
Building configuration...
[OK]
```

```
SWC>ena
SWC#conf t
Enter configuration commands, one per line. End with CNTL/Z.
SWC(config)#interface range f0/23 - 24
SWC(config-if-range)#channel-group 3 mode active
%EC-5-L3DONTBNDL2: Fa0/23 suspended: LACP currently not enabled on the remote port.
%EC-5-L3DONTBNDL2: Fa0/24 suspended: LACP currently not enabled on the remote port.
SWC(config-if-range)#
Creating a port-channel interface Port-channel 3

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINK-5-CHANGED: Interface Port-channel3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel3, changed state to up

SWC(config-if-range)#exit
SWC(config)#interface port-channel 3
SWC(config-if)#switchp
SWC(config-if)#switchport mode trunk
SWC(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/23, changed state to up
%LINK-3-UPDOWN: Interface Port-channel3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel3, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/24, changed state to up
%LINK-5-CHANGED: Interface Port-channel3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Port-channel3, changed state to up

SWC(config-if)#end
SWC#
%SYS-5-CONFIG_I: Configured from console by console

SWC#write memory
Building configuration...
[OK]
SWC#
```